

选择题

1. 无失真选项

2. B

计算题

1. 解: $H(\omega) = -j\omega = 2\omega e^{-j\frac{\pi}{2}}$

$$|H(\omega)| = 2|\omega|$$

$$\angle H(\omega) = -\frac{\pi}{2}$$

1) $x(t) = e^{jt} \quad \omega_0 = 1$

$$y(t) = H(\omega_0) e^{jt} = 2 e^{j(t-\frac{\pi}{2})} = -2je^{jt}$$

2) $x(t) = (\sin \omega_0 t) u(t)$

$$t \geq 0 \text{ 时 } x(t) = \sin \omega_0 t = \cos(\frac{\pi}{2} - \omega_0 t + \frac{\pi}{2})$$

$$y(t) = |H(\omega_0)| \cos(-\omega_0 t + \frac{\pi}{2} + \angle H(\omega_0))$$

$$= 2\omega_0 \cos(-\omega_0 t) = -2\omega_0 \cos(\omega_0 t)$$

$$t < 0 \text{ 时 } x(t) = 0 \quad y(t) = 0$$

$$\text{则 } y(t) = -2\omega_0 \cos(\omega_0 t) u(t)$$

3) $X(\omega) = \frac{1}{j\omega(b+j\omega)}$

$$Y(\omega) = H(\omega) X(\omega) = \frac{2\omega e^{-j\frac{\pi}{2}}}{j\omega(b+j\omega)} = -\frac{2}{b+j\omega}$$

$$= \frac{-12 + j2\omega}{36 + \omega^2}$$

4) $X(\omega) = \frac{1}{2+j\omega}$

$$Y(\omega) = H(\omega) X(\omega) = \frac{-j2\omega}{2+j\omega} = \frac{-2\omega^2 - j4\omega}{4 + \omega^2}$$

2. 解: 通频带: $|\omega| \leq 3\pi$

$$H(\omega) = \begin{cases} \frac{\omega}{3\pi} e^{j\frac{\pi}{2}} & \omega > 0 \\ \frac{-\omega}{3\pi} e^{j\frac{\pi}{2}} & \omega < 0 \end{cases}$$

(1) $x(t) = \cos(2\pi t + \theta)$ ~~$\omega = 2\pi$~~ $\omega = 2\pi > 0$ 且 $\omega < 3\pi$

$$y(t) = |H(\omega)| \cos(2\pi t + \theta + \angle H(\omega))$$

$$= \frac{2}{3} \cos(2\pi t + \theta + \frac{\pi}{2}) = -\frac{2}{3} \sin(2\pi t + \theta)$$

(2) $x(t) = \cos(4\pi t + \theta)$ $\omega = 4\pi > 3\pi$

$$y(t) = 0$$

(3) $m \leq t \leq (m + \frac{1}{2})$ 时

$$x(t) = \sin 2\pi t = \cos(-2\pi t + \frac{\pi}{2}) \quad \omega = 2\pi > 0 \quad \text{且} \quad \omega < 3\pi$$

$$y(t) = |H(\omega)| \cos(-2\pi t + \frac{\pi}{2} + \angle H(\omega)) = \frac{2}{3} \cos(-2\pi t + \pi)$$

$$= -\frac{2}{3} \cos 2\pi t$$

$(m + \frac{1}{2}) \leq t \leq (m+1)$ 时

$$x(t) = 0 \quad y(t) = 0$$

$$\text{则} \quad y(t) = \begin{cases} -\frac{2}{3} \cos 2\pi t & m \leq t \leq (m + \frac{1}{2}) \\ 0 & (m + \frac{1}{2}) \leq t \leq (m+1) \end{cases}$$

3. 解: $H(\Omega)$ 、 $x(n)$ 均为周期性

即

$$H(\Omega) = \begin{cases} \frac{\Omega}{\pi} e^{j\frac{\Omega}{2}} & 0 \leq \Omega \leq \pi \\ -\frac{\Omega}{\pi} e^{-j\frac{\Omega}{2}} & -\pi \leq \Omega < 0 \end{cases} = j\frac{\Omega}{\pi}$$

$$\text{即 } |H(\Omega)| = \frac{|\Omega|}{\pi} \quad \angle H(\Omega) = \begin{cases} \frac{\pi}{2} & 0 < \Omega \leq \pi \\ -\frac{\pi}{2} & -\pi \leq \Omega < 0 \end{cases}$$

周期为 2π

$$x(n) = \cos(\Omega_0 n + \theta) \quad \text{不妨令 } 0 < \Omega_0 \leq \pi$$

~~$$y(n) = |H(\Omega_0)| \cos(\Omega_0 n + \theta + \angle H(\Omega_0))$$~~

~~$$x(n) = \frac{1}{2} e^{j(\Omega_0 n + \theta)} + \frac{1}{2} e^{-j(\Omega_0 n + \theta)}$$~~

~~$$y(n) = H(\Omega_0) x(n) = \frac{1}{2} \frac{\Omega_0}{\pi} e^{j(\Omega_0 n + \theta + \frac{\pi}{2})} + \frac{1}{2} \frac{\Omega_0}{\pi} e^{-j(\Omega_0 n + \theta)}$$~~

~~$$y(n) = \frac{1}{2} H(\Omega_0) e^{j(\Omega_0 n + \theta)} + \frac{1}{2} H(-\Omega_0) e^{j(-\Omega_0 n - \theta)}$$~~

~~$$= \frac{\Omega_0}{2\pi} e^{j(\Omega_0 n + \theta + \frac{\pi}{2})} + \frac{\Omega_0}{2\pi} e^{j(-\Omega_0 n - \theta - \frac{\pi}{2})}$$~~

~~$$= \frac{\Omega_0}{2\pi} (j e^{j(\Omega_0 n + \theta)} + j e^{-j(\Omega_0 n + \theta)})$$~~

~~$$= \frac{j\Omega_0}{2\pi} (e^{j(\Omega_0 n + \theta)} + e^{-j(\Omega_0 n + \theta)})$$~~

~~$$= j \frac{\Omega_0}{2\pi} \cos(\Omega_0 n + \theta)$$~~

$$y(n) = H(\Omega_0) \frac{1}{2} e^{j(\Omega_0 n + \theta)} + \frac{1}{2} H(-\Omega_0) \frac{1}{2} e^{j(-\Omega_0 n - \theta)}$$

$$= \frac{\Omega_0}{2\pi} e^{j(\Omega_0 n + \theta + \frac{\pi}{2})} + \frac{\Omega_0}{2\pi} e^{j(-\Omega_0 n - \theta - \frac{\pi}{2})}$$

$$= \frac{\Omega_0}{2\pi} (j e^{j(\Omega_0 n + \theta)} - j e^{-j(\Omega_0 n + \theta)})$$

$$= -\frac{\Omega_0}{\pi} \sin(\Omega_0 n + \theta)$$